MAT 3007 Homework # 8 Spring 2024

Due: Monday, May 13 at 12:00pm (noon)

Please note: You should submit your answers (scanned or typeset) in a PDF file. Questions 2, 3 and 5 ask you to write codes in Python or MATLAB. Submit all necessary code files via Blackboard. Also, give meaningful names to your decision variables and parameters, and add comments to your code liberally. The problem set is due at 12:00pm (noon) on Monday, May 13th.

1. (20 pts) Coach Dai is trying to choose the starting lineup for the CUHK-SZ men's basketball team for the opening game. The team consists of seven players who have been rated (on a scale of 1 = poor to 3 = excellent) according to their shooting ability. The positions that each player is able to play and the player's shooting ability is listed below:

Player	Position	Shooting
1	Guard	3
2	Center	1
3	Guard & Forward	3
4	Forward & Center	3
5	Guard & Forward	1
6	Forward	2
7	Guard & Forward	2

The five-player starting lineup must satisfy the following restrictions. Please define proper decision variables, and help Coach Dai to write down the data-specific constraint corresponding to each restriction using your defined decision variables.

(a) The starting lineup has five players;

(b) At least 2 players must be able to play forward;

(c) The average shooting ability must be at least 2;

(d) Only one of player 3 and player 6 can start;

(e) Either player 2 or player 3 must start (possible that both of them will start).

2. (20 pts) Three different products are made on three production lines each week. If a production line is used in a given week, there is an associated setup cost. Each worker is designated to only one production line, and the pay and production of each worker depends on which line they are assigned to. In addition, each worker is assigned to one specific product on their assigned line. Relevant data are given in the following table.

	Line 1	Line 2	Line 3
Setup cost	\$2000	\$3000	\$4000
Units of product 1/worker	50	90	120
Units of product 2/worker	75	110	130
Units of product 3/worker	90	125	150
Cost/worker	\$700	\$1000	\$1500

This week, we need to make 600 units of product 1, 800 units of product 2, and 1000 units of product 3. At most 20 workers are available this week.

(a) Write down an integer linear programming model that minimizes the total cost of meeting this week's demand. As usual, be clear about what your variables represent.

(b) Implement and solve your model using CVX. Report your solution "by hand" in a way that someone without optimization background can understand.

For MATLAB, to solve the integer programming, you need to first load the solver (e.g., Gurobi):

```
cvx_solver gurobi
```

You may define the binary/integer variables as follows:

```
variable x binary
variable y integer
```

For Python, you may define the binary/integer variables as follows:

```
import cvxpy as cp
x = cp.Variable(n, boolean=True)
y = cp.Variable(n, integer=True)
```

When you solve the problem, you can load your favorite solver (e.g., COPT) with:

```
prob = cp.Problem(objective, constraints)
prob.solve(solver="COPT")
```

3. (20 pts) To graduate from FastTrack University with a major in operations research, a student must complete at least two math courses, at least two OR courses, and at least two computer courses. Some courses can be used to fulfill more than one requirement (shown by "X"), and some courses are prerequisites for others. This information is given in the following table.

	Fulfill requirements for			
Courses	Math	OR	Computer	Prerequisite
1. Calculus	X			
2. Operations Research	X	X		
3. Data Structures	X		X	Intro to Programming
4. Business Statistics	X	X		Calculus
5. Computer Simulation		X	X	Intro to Programming
6. Intro to Programming			X	
7. Forecasting	X	X		Business Statistics

(a) Formulate and write down a linear integer programming that minimizes the number of courses needed to satisfy the major requirements.

(b) Implement and solve your model using CVX. Report your solution "by hand" in a way that someone without optimization background can understand.

4. (20 pts) Consider the integer linear program:

$$\begin{aligned} \min \quad & 16x_1 + 14x_2 + 15x_3 && (1)\\ \text{s.t.} \quad & x_1 + x_2 \geq 1 && (2)\\ & x_2 + x_3 \geq 1 && (3)\\ & x_3 + x_1 \geq 1 && (4)\\ & x_1, x_2, x_3 \in \{0, 1\}. && (5) \end{aligned}$$

(a) Find the optimal solution by inspection or enumeration.

(b) Show that an equivalent integer linear program would result if the main constraints (2)-(4) were replaced by

$$x_1 + x_2 + x_3 \geq 2,$$

i.e., that the feasible sets of the two problems are identical.

(c) Compare the optimal solutions of the linear programming relaxation of the two formulations. Discuss what you observe and which formulation is better. You can use CVX to solve the linear programming relaxation problems but you do not have to upload your codes.

5. (20 pts) Solve the following MILP with branch-and-bound. You may choose any reasonable branching variable selection and tree search strategy you want. Follow the same instructions as in lecture slides to illustrate your steps of the method. Solve the LPs for the nodes with MATLAB/Python. You need to submit your MATLAB/Python codes on BB.

$$\begin{aligned}
\max \quad & 5x_1 + 3x_2 + 4x_3 \\
\text{s.t.} \quad & 43x_1 + 25x_2 + 57x_3 \leq 150 \\
& 83x_1 + 34x_2 + 63x_3 \leq 200 \\
& 0 \leq x_i \leq 2 \qquad \forall i = 1, 2, 3 \\
& x_i \in \mathbb{Z} \qquad \forall i = 1, 2, 3
\end{aligned}$$